Chhavi Singh

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Education

- **Dr. B. R. Ambedkar National Institute of Technology, Jalandhar** — Punjab, India
  B.Tech. in Instrumentation and Control Engineering — CGPA: 8.18 — *July 2022 – June 2026*
  Courses: *Modern Control Systems, Industrial Automation, Process Dynamics and Control*
- **St. Joseph School, Shaktinagar (CBSE)** — Uttar Pradesh, India
  Class XII - PCM; Percentage: 94.8% — *2021*
- **St. Joseph School, Shaktinagar (CBSE)** — Uttar Pradesh, India
  Class X; Percentage: 97% — *2019*

Skills Summary

- **Software & Tools**: MATLAB, ROS2, LabVIEW, Python
- **Industrial Automation**: PLC Programming (RLL), Robust Control, SCADA Concepts
- **Instrumentation**: Sensors & Transmitters, DAQ Systems, Vibration Measurement

Experience

- **Indian Institute of technology Madras – DiRo Lab** — Chennai, TN
  Project Associate – Mechatronics & Control — *July 2026 – October 2026*
  - **Remote Controlled Weapon Station Stabilization (Simscape Demo)**: Developed a cascaded PI + SMC-ADRC control architecture for a 7.62 mm remote controlled weapon station, achieving sub-mrad line-of-sight (LOS) tracking accuracy. The project was undertaken in partnership with Electro Pneumatics and Hydraulics Pvt. Ltd.
- **Indian Institute of Technology Kanpur – SMSS Lab** — Kanpur, UP
  Research Intern — Robotics & Control — *January 2026 – June 2026*
  - **Robust Trajectory Tracking Controllers for Skid-Steer Autonomous UGV (Demo)**: B.Tech thesis on robust trajectory tracking of skid-steer UGVs. Developed a high-fidelity Simscape Multibody model incorporating nonlinear wheel–ground interaction, slip dynamics, and terrain disturbances. Designed and evaluated Lyapunov + SMC + STA, adaptive Backstepping + SMC + STA, and NMPC + ADRC + LESO architectures for circular and discontinuous trajectories, demonstrating robust tracking and disturbance rejection. (Presentation)
- **Northern Coalfields Limited (NCL), Coal India Ltd.** — Singrauli, MP
  Vocational Trainee — Automation & Instrumentation — *June 2025 – July 2025*
  - Developed a detailed **PLC automation framework** for the Coal Handling Plant (CHP), covering conveyor sequencing, crusher feed logic, stacker–reclaimer operation, and motor interlocking.
  - Modelled **ladder logic blocks** for start/stop sequences, zero-speed and belt-sway protection, pull-cord trip, and upstream–downstream permissive logic.

Publications

- **Robust Control Architectures for Trajectory Tracking of Autonomous Skid-Steer UGVs** — 2026
  Conference: IECON 2026 *Status: Accepted*
- **Real-Time Non-Contact Soil Moisture Estimation Using Machine Learning at the Edge** — Dec 2025
  Conference: IEEE ICEdge 2025, IISc Bangalore *Status: presented* *link*

Honors and Awards

- **Awarded the IES SYPA grant as one of the top 52 papers selected for IECON 2026 (Video)**
- Awarded the Student Travel Grant for ICEdge 2025
- Awarded the NTPC Scholarship for excellent performance in class X boards - May, 2019
- 1st Prize in Painting Competitions – Utkansh, Annual Cultural Fest, NIT Jalandhar

Volunteer Experience

- Member — National Service Scheme (NSS) Jalandhar, India Participated in local surveys, awareness campaigns, and community outreach eventsNov 2022 - Nov 2023
- D-SAMP Mentor for Aug 2024-Aug 2025
- Core Member at LADC (Literature and Debate Club) and FAS (Fine Arts Society)